



University of Piraeus

School of Information and Telecommunications Technologies

Department of Digital Systems

Level: Postgraduate Program of Study

Network Security

Case 5

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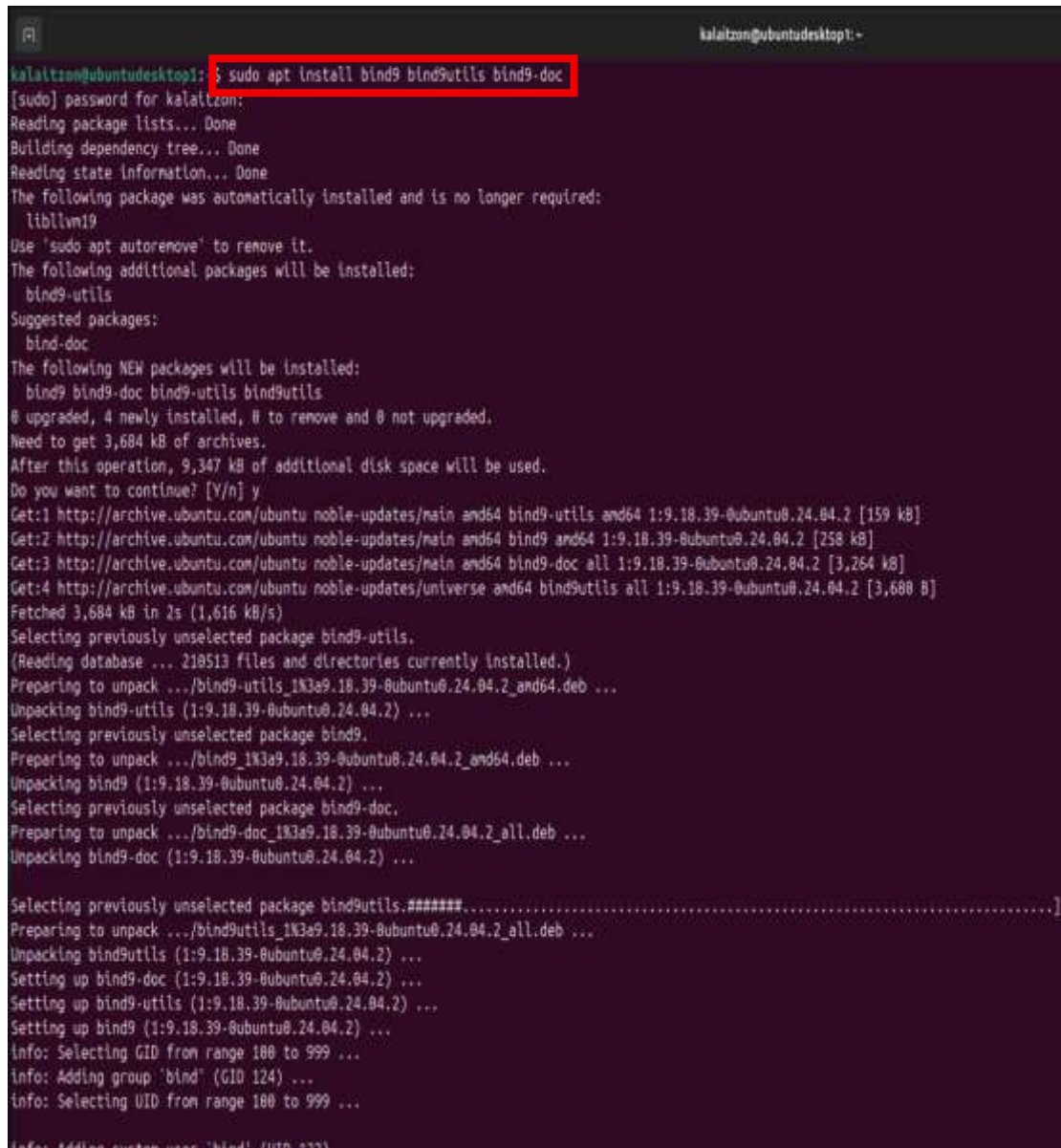
Abstract

Install DNS server using the BIND9 tool on Ubuntu and create a Master Zone for the domain "Kalaitzon.gr". Set up the Forwarding to Google function and connect the domain to the server's IP. The correct operation was confirmed through a check by Windows 10 (client).

Contents

Practical piece of work.....	1
Bibliography	3

Practical piece of work



```
kalaitzon@ubuntu desktop1: ~$ sudo apt install bind9 bind9utils bind9-doc
[sudo] password for kalaitzon:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done

The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  bind9-utils
Suggested packages:
  bind-doc
The following NEW packages will be installed:
  bind9 bind9-doc bind9-utils bind9utils
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 3,684 kB of archives.
After this operation, 9,347 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-utils amd64 1:9.18.39-0ubuntu0.24.04.2 [159 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9 amd64 1:9.18.39-0ubuntu0.24.04.2 [258 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-doc all 1:9.18.39-0ubuntu0.24.04.2 [3,264 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 bind9utils amd64 1:9.18.39-0ubuntu0.24.04.2 [3,680 B]
Fetched 3,684 kB in 2s (1,616 kB/s)
Selecting previously unselected package bind9-utils.
(Reading database ... 218513 files and directories currently installed.)
Preparing to unpack .../bind9-utils_1:9.18.39-0ubuntu0.24.04.2_amd64.deb ...
Unpacking bind9-utils (1:9.18.39-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_1:9.18.39-0ubuntu0.24.04.2_amd64.deb ...
Unpacking bind9 (1:9.18.39-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9-doc.
Preparing to unpack .../bind9-doc_1:9.18.39-0ubuntu0.24.04.2_all.deb ...
Unpacking bind9-doc (1:9.18.39-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9utils.#####
Preparing to unpack .../bind9utils_1:9.18.39-0ubuntu0.24.04.2_all.deb ...
Unpacking bind9utils (1:9.18.39-0ubuntu0.24.04.2) ...
Setting up bind9-doc (1:9.18.39-0ubuntu0.24.04.2) ...
Setting up bind9-utils (1:9.18.39-0ubuntu0.24.04.2) ...
Setting up bind9 (1:9.18.39-0ubuntu0.24.04.2) ...
info: Selecting GID from range 180 to 999 ...
info: Adding group 'bind' (GID 124) ...
info: Selecting UID from range 180 to 999 ...
info: Adding system user 'bind' (UID 372)
```

We "download" bind9 with the command **sudo apt install bind9 bind9utils bind9-doc**.

Then we set the Forwarding (Recursive Query). Without it, our server can only answer queries for our own domain (e.g. Kalaitzon.gr). With the command **sudo nano /etc/bind/named.conf.options** we open the settings file and remove the comment (//) by setting the Google IP (8.8.8.8 which is normally 008.008.008.008). We save it and exit by pressing Ctrl + O and Enter. Then Ctrl + X to return to the terminal. We do this

because when the user, e.g. from Windows, asks the server for example what the IP of YouTube is, it will transfer the question to the IP of Google and in this way the server will give the answer. Under other circumstances our server would be isolated and would not know anything other than our own domain.

```
GNU nano 7.2
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk. See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    // forwarders {
    //     0.0.0.0;
    // };

    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys. See https://www.isc.org/bind-keys
    //=====
    dnssec-validation auto;

    listen-on-v6 { any; };
};
```



```
GNU nano 7.2
options {
    directory "/var/cache/bind";

    // If there is a firewall between you and nameservers you want
    // to talk to, you may need to fix the firewall to allow multiple
    // ports to talk. See http://www.kb.cert.org/vuls/id/800113

    // If your ISP provided one or more IP addresses for stable
    // nameservers, you probably want to use them as forwarders.
    // Uncomment the following block, and insert the addresses replacing
    // the all-0's placeholder.

    forwarders {
        8.8.8.8;
    };

    //=====
    // If BIND logs error messages about the root key being expired,
    // you will need to update your keys. See https://www.isc.org/bind-keys
    //=====
    dnssec-validation auto;

    listen-on-v6 { any; };
};
```

Then we will find the IP of the host with the command **ip a**. The IP of Ubuntu is **192.168.137.129**

```
kalaitzon@ubuntudesktop1:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens32: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:f5:51:1f brd ff:ff:ff:ff:ff:ff
    altname enp2s0
    inet 192.168.137.129/24 brd 192.168.137.255 scope global dynamic noprefixroute ens32
        valid_lft 1223sec preferred_lft 1223sec
    inet6 fe80::20c:29ff:fef5:511f/64 scope link
        valid_lft forever preferred_lft forever
```

Next, we proceed to create our own Local Zone. With the command **sudo nano /etc/bind/named.conf.local** we edit the configuration file to declare the domain "Kalaitzon.gr". Within the file, we define the zone type as master, thus declaring that our server will be the Authoritative Name Server for the specific domain. The creation of the Local Zone is necessary because the domain "Kalaitzon.gr" It is private and does not exist on the internet. Therefore, DNS servers, such as Google's, do not have records for this and are unable to resolve it. Through the local zone, we define our own server as the Authoritative Source for that domain. Thus, we achieve the resolution of names locally, without depending on external providers, essentially creating an internal network. At the same time, with the file command, we connect the zone to the db.Kalaitzon.gr file (which we will create below). This will be the data file that will

include all the necessary information of our domain. In other words, we separate the resolution process. If a client requests an address outside of our domain, the Forwarding we previously set up is activated. On the

```
GNU nano 7.2
//
// Do any local configuration here
//
// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

zone "Kalaitzon.gr" {
    type master;
    file "/etc/bind/db.Kalaitzon.gr";
};
```

contrary, for queries related to "Kalaitzon.gr", the server does not address the Google IP, but draws the information directly from the specific local file.

Now we will create the file that we declared in the zone above. With the command **sudo cp /etc/bind/db.local /etc/bind/db. Kalaitzon.gr** we will copy the template file that exists from localhost to ours (db.Kalaitzon.gr) and then edit it with the **sudo nano /etc/bind/db command. Kalaitzon.gr**. Where it has localhost, we replace it with Kalaitzon.gr (our domain) and where we see IP we put the one of Ubuntu (192.168.137.129).

```
kalaitzon@ubuntu desktop1: ~  
GNU nano 7.2 /etc/bind/db.Kalaitzon.gr  
; BIND data file for local loopback interface  
;  
$TTL 604800  
@ IN SOA localhost. root.localhost. (  
    2      ; serial  
    604800 ; Refresh  
    86400  ; Retry  
    2419200 ; Expire  
    604800 ) ; Negative Cache TTL  
;  
@ IN NS localhost.  
@ IN A 127.0.0.1  
@ IN AAAA ::1
```

```
kalaitzon@ubuntu desktop1: ~  
GNU nano 7.2 /etc/bind/db.Kalaitzon.gr  
; BIND data file for local loopback interface  
;  
$TTL 604800  
@ IN SOA Kalaitzon.gr. root.Kalaitzon.gr. (  
    2      ; serial  
    604800 ; Refresh  
    86400  ; Retry  
    2419200 ; Expire  
    604800 ) ; Negative Cache TTL  
;  
@ IN NS ns.Kalaitzon.gr.  
@ IN A 192.168.137.129  
ns IN A 192.168.137.129  
www IN A 192.168.137.129
```

```
kalaitzon@ubuntu desktop1: ~  
kalaitzon@ubuntu desktop1: ~  
● named.service - BIND Domain Name Server  
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)  
   Active: active (running) since Sat 2025-11-22 01:38:52 EET; 3s ago  
     Docs: man:named(8)  
   Main PID: 13461 (named)  
    Status: "running"  
   Tasks: 10 (limit: 9374)  
  Memory: 23.9M (peak: 24.4M)  
     CPU: 297ms  
   CGroup: /system.slice/named.service  
           └─13461 /usr/sbin/named -f -u bind  
  
Nov 22 01:38:54 ubuntu desktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#53  
Nov 22 01:38:54 ubuntu desktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:503:ba3e::2:30#53  
Nov 22 01:38:54 ubuntu desktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:503:c27::2:30#53  
Nov 22 01:38:54 ubuntu desktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:500:2::c#53  
Nov 22 01:38:54 ubuntu desktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:7fd::1#53
```


We do the standard typing to save the file and proceed to restart the server with the command **sudo systemctl restart bind9**. If it does not get an error message, then it means that it worked. To check that everything went well and it works, we will use the command **sudo systemctl status bind9** and we will observe the line that says **Active: active (running)**

To confirm the proper functioning of the DNS Server, a second virtual machine with Windows 10 operating system was used, which acted as a Client. Initially, an attempt was made to resolve the domain name `www.kalaitzon.gr` through the `nslookup` command. This attempt failed (displaying a "Non-existent domain" message) because the client was configured to use the network's default DNS server (VMware Gateway, 192.168.137.2), who did not have registrations for our local zone. To address this, the client's Network Adapter Settings have been modified. Specifically, in the TCP/IPv4 protocol, the IP address of our Ubuntu Server (192.168.137.129) was statically set as Preferred DNS Server. After applying the settings, the **nslookup command `www.kalaitzon.gr`** executed again. This time, the client contacted our DNS Server directly, which responded successfully, resolving the domain name to the correct IP

address (192.168.137.129), thus confirming that the Authoritative Server has been configured correctly.

```
C:\> Command Prompt

Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

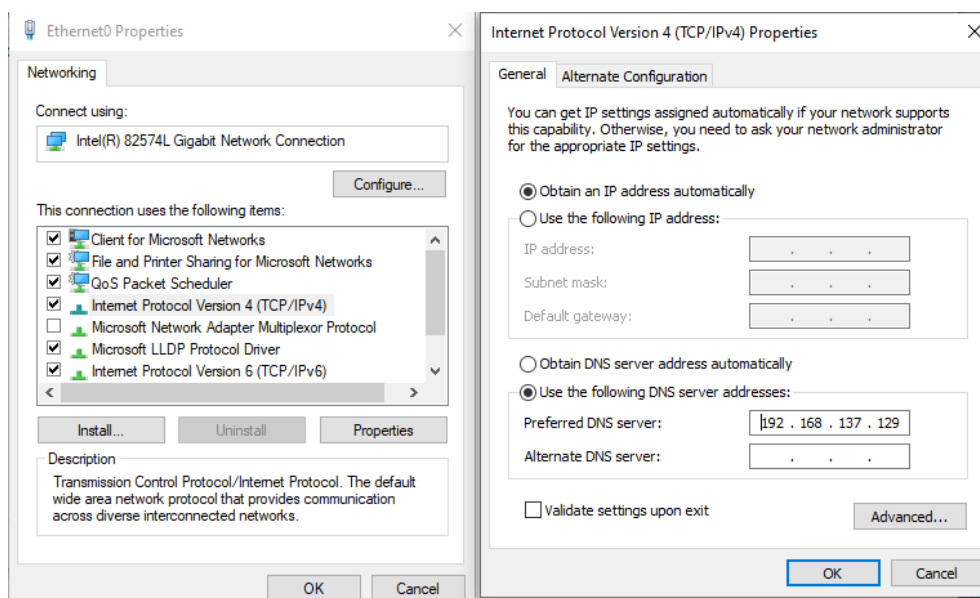
C:\Users\Client_Win10>nslookup www.kalaitzon.gr
Server: UnKnown
Address: 192.168.137.2

*** UnKnown can't find www.kalaitzon.gr: Non-existent domain

C:\Users\Client_Win10>nslookup www.kalaitzon.gr
Server: UnKnown
Address: 192.168.137.129

Name: www.Kalaitzon.gr
Address: 192.168.137.129

C:\Users\Client_Win10>
```



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